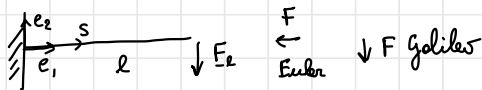


Euler's elastica and Galileo's beam problem



Const. eqs $\epsilon = 0$, $M = EI k = B k = B \alpha'$

kin BC's $\underline{z}(0) = 0$, $\alpha(0) = 0$ (clump at $s=0$)

$$\text{Loads } \underline{f} = 0, m = 0, M_k = 0, \underline{F}_k = F_1 \underline{e}_1 + F_2 \underline{e}_2 \quad \left\{ \begin{array}{l} \text{Euler } F_2 = 0, F_1 = -F \\ \text{Galileo } F_1 = 0, F_2 = -F \end{array} \right.$$

Equil. eqs ($s = s_F$)

$$\underline{R}' = 0 \Rightarrow \underline{R}(s) = \underline{R}(l) = F_1 \underline{e}_1 + F_2 \underline{e}_2 = N \underline{t} + T \underline{m}$$

$$M' + T = 0$$

$$\Rightarrow T = \underline{R} \cdot \underline{m} = \begin{cases} -F \underline{e}_1 \cdot (-\sin \alpha, \cos \alpha) = F \sin \alpha & \text{Euler} \\ -F \underline{e}_2 \cdot (-\sin \alpha, \cos \alpha) = -F \cos \alpha & \text{Galileo} \end{cases}$$

• Reduced problem of elastica (Euler)

$$\begin{cases} B \alpha''(s) + F \sin \alpha(s) = 0 & \text{2nd order ODE in } \alpha. \text{ Once solved, } \underline{x}(s) = \int_0^s \cos \alpha(\sigma) d\sigma \underline{e}_1 + \\ \alpha(0) = 0 & \\ M(l) = B \alpha'(l) = 0 \Rightarrow \alpha'(l) = 0 & + \int_0^s \sin \alpha(\sigma) d\sigma \underline{e}_2 \end{cases}$$

$\alpha = 0$ solution $\forall F \rightarrow$ can study stability with small angle expansion around $\alpha = 0$ ($\sin \alpha \approx \alpha$)

$$B \alpha'' + F \alpha = 0 \Rightarrow \alpha'' + \frac{F}{B} \alpha = 0 \quad (\text{harmonic oscillator})$$

$$\Rightarrow \alpha(s) = a \cos\left(\sqrt{\frac{F}{B}} s\right) + b \sin\left(\sqrt{\frac{F}{B}} s\right)$$

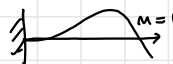
$$\alpha(0) = 0 \Rightarrow a = 0 \quad \alpha(s) = b \sin\left(\sqrt{\frac{F}{B}} s\right)$$

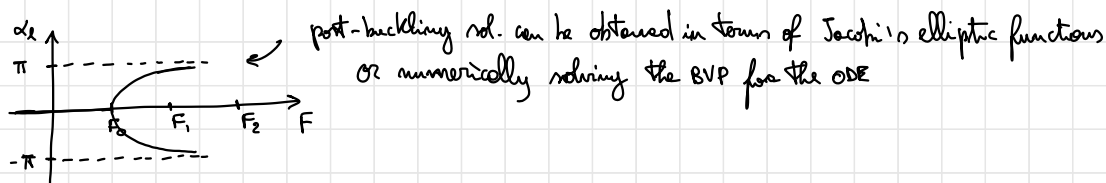
$$\alpha'(l) = b \sqrt{\frac{F}{B}} \cos\left(\sqrt{\frac{F}{B}} l\right) = 0 \Rightarrow \sqrt{\frac{F}{B}} l = \frac{\pi}{2} + m\pi \quad m = 0, 1, \dots$$

$$\sqrt{\frac{F_m}{B}} l = \frac{\pi}{2} (1 + 2m) \Rightarrow F_m = \frac{\pi^2 B}{4 l^2} (1 + 2m)^2$$

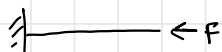
$$m = 0 \text{ (smallest } m \text{ for which } \alpha \neq 0 \text{ possible)}, F_0 = F_{\text{crit}} = \frac{\pi^2 B}{4 l^2}, \alpha_0(s) = b \sin\left(\frac{\pi}{2} \frac{s}{l}\right) \quad b \text{ arbitrary}$$

$$\text{More generally, } \alpha_m(s) = \alpha_l \sin\left(\frac{\pi}{2} (1 + 2m) \frac{s}{l}\right)$$



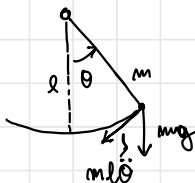


The elastica eqn is a famous one (Kirchhoff's kinetic analogy)



$$\alpha'' + \frac{F}{B} \sin \alpha = 0$$

$$\alpha(0) = 0, \alpha'(l) = 0$$



$$ml\ddot{\theta} + mg \sin \theta = 0 \Rightarrow \ddot{\theta} + \frac{g}{l} \sin \theta = 0$$

$$t \in (0, T) \quad \theta(0) = 0, \dot{\theta}(T) = 0$$

We will solve both Euler's and Galileo's problems numerically.

In conclusion

One of the key concepts of this course is that the principle of virtual powers

$$\int_0^l H \dot{K} + N \dot{E} = \int_0^l \underline{f} \cdot \underline{\dot{x}} + m \dot{\omega} \quad (*)$$

is the weak form of the differential equations of equilibrium

$$\begin{cases} \underline{R}' + \underline{f} = 0 \\ M' + T + m = 0 \end{cases} \quad (**)$$

This is general and for reaching, beyond mechanics, to any system governed by differential equations.

In the special case of elastic beams subject to conservative loads, PPV has an energetic interpretation as the principle of stationary potential energy

$$V(\underline{x}) = V_{el}(\underline{x}) - W_{ext}(\underline{x})$$

$$\left. \frac{d}{dh} V(\underline{x} + h \delta \underline{x}) \right|_{h=0} = \underbrace{\frac{\delta V}{\delta \underline{x}}}_{\delta \underline{x}} = W_{el}(\delta \underline{x}) - W_{ext}(\delta \underline{x}) = 0 \quad (***)$$

physicists call this variational derivative of the functional V with respect to the function $s \mapsto \underline{x}(s)$; in Calculus of Variations this is called the first variation of the functional V $(***)$ is exactly $(*)$, which is the weak form of $(**)$. The most widely spread method to solve $(**)$ numerically is based on $(*)$, and is called the Finite Element Method.